

Adaptive Fabric Surface Defect Detection and Localization using Classical Texture Descriptors

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I. WORK DONE THIS WEEK:-

This week we mainly worked on understanding our project topic and deciding how we wanted to approach the problem. Our group decided to work on **Adaptive Fabric Surface Defect Detection and Localization using Classical Texture Descriptors**. We found this topic interesting because fabric normally has a repetitive texture, and a defect in the fabric can disturb this regular pattern. We wanted to see whether these changes can be detected using the Digital Image Processing techniques that we have been learning instead of directly depending on more advanced deep learning methods.

We first tried to understand what exactly we need to achieve through the project. We felt that just identifying whether an image has a defect would not be enough for our project, because the word *localization* is also important in our topic. So, our final objective would be to detect a defect and also find approximately where the defect is present in the fabric image.

We looked at some sample fabric images and started thinking about the preprocessing that would be required before applying any texture-based method. Depending on the dataset, we may have to convert the images to grayscale, resize them, remove unwanted noise, and normalize the images. We have not finalized all of these steps yet because we first want to understand the dataset properly and see what kind of images and defects we are dealing with.

After this, we started looking into different classical texture descriptors. The two methods that seemed most relevant to us were **Gray-Level Co-occurrence Matrix (GLCM)** and **Local Binary Pattern (LBP)**. From our initial understanding, GLCM gives us statistical information about how different gray levels occur next to each other, while LBP looks at local patterns around individual pixels. Since fabric has a lot of texture information, we felt that both of these methods could be useful for our problem.

We also discussed the meaning of the word **adaptive** in our project. A fixed threshold may not work equally well for every fabric image because the texture and brightness can change from one image to another. Because of this, we are planning to investigate an adaptive way of deciding whether a particular region is different enough from the normal fabric texture to be considered a defect.

For localization, one idea we discussed was dividing the image into smaller windows or patches and calculating texture information for each region. If one region is significantly different from the surrounding normal texture, it could indicate the presence of a defect. We will explore this idea further once we start implementing the basic preprocessing and texture extraction.

Our initial understanding of the project can therefore be represented as: **Fabric Image → Preprocessing → Texture Analysis → Defect Detection → Defect Localization**.

At this point, we are mainly trying to understand the problem and the techniques before starting the main implementation.

II. WORK TO BE DONE NEXT WEEK:-

Next week, we plan to start working on the coding part of the project. Our first step will be to properly inspect and organize the dataset that we are going to use.

We plan to start with the basic preprocessing of the images. We will try converting the images to grayscale and check whether any noise is present in the images. If required, we will experiment with basic filtering techniques that we have already learned in the course. We will also look at whether the images need resizing or normalization so that they can be processed consistently.

After implementing the basic preprocessing, we will compare the original and processed images to understand whether our preprocessing is actually helping or not. We do not want to apply filters unnecessarily if they end up removing useful texture information.

We will also start preparing the code for extracting texture features using GLCM and LBP. At the same time, we plan to continue reading a few more papers to understand how other researchers have approached fabric-defect detection and which techniques can realistically be implemented using the concepts we have learned in Digital Image Processing.

From next week onwards, we will also maintain the code and results in our GitHub repository along with the weekly progress report so that our progress can be tracked regularly.

III. LITERATURE REVIEW:-

During this week, we started looking into research papers related to fabric-defect detection and classical texture analysis. Since we are still at the beginning of the project, our main purpose was to understand the basic ideas used by other researchers rather than trying to directly reproduce their complete methods.

One of the papers we looked at was *Fabric Defect Detection Using Modified Local Binary Patterns*. The paper uses Local Binary Patterns for identifying defects in fabric by looking at local texture information. We found this useful because LBP is one of the techniques we are planning to investigate. The idea of comparing local regions of fabric also relates to our plan of eventually dividing the image into smaller windows for localization.

We also looked at *Fabric Defect Detection based on GLCM Approach*. This paper uses the Gray-Level Co-occurrence Matrix to obtain texture information from fabric images. From this paper, we got a better idea of why statistical texture features such as contrast, homogeneity, energy and correlation can be useful when trying to distinguish normal fabric from defective fabric.

Another paper we looked at was *Fabric Defect Classification Based on LBP and GLCM*. This was particularly interesting to us because it combines the two techniques that we are currently planning to study. LBP provides information about local texture patterns, while GLCM provides statistical information about the texture. We would like to test both separately first and then see whether combining them gives better results.

We also came across work involving local/window-based analysis and adaptive approaches for fabric-defect detection. This made us consider that using one fixed threshold for every image may not be the best approach. We therefore want to investigate an adaptive method later in the project rather than deciding on a fixed threshold from the beginning.

At this stage, the literature review has mainly helped us understand the problem and decide which techniques are worth trying first. We plan to go through more papers as we start implementing the project and have a clearer idea of the dataset and the type of defects we are dealing with.

IV. CHALLENGES FACED:-

The main challenge we faced this week was understanding how to distinguish actual defects from normal variations in fabric texture. Fabric already contains a lot of repeated patterns, so not every change in the image will necessarily be a defect. We also need to decide how to divide the image into suitable regions without losing important texture information. Another point that we still need to explore is how to choose an adaptive threshold that works across different fabric images.

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